

Memi 2.7.3 Confirmatory Audit and 2.7.4 Fail-Closed Release Assessment

Technical audit package · V15 frozen confirmatory protocol

1 August 2026

Artifact status: INCOMPLETE — GENERATED RESULTS NOT LOADED

No confirmatory quality, efficiency, or superiority claim is authorized while generated analysis artifacts are absent.

Abstract

This audit reconstructs the exact `@memi-design/cli@2.7.3` release, preserves the earlier V14 evidence under its actual 2.7.2 label, and evaluates the exact published 2.7.3 tarball under a frozen, three-platform, 18-pair protocol. The primary estimand is the within-task paired difference in a blinded, model-graded design-quality score. Functional acceptance and resource use are secondary outcomes. The associated 2.7.4 engineering change replaces permissive skill-fitness reuse with exact route identity, immutable quality evidence, chronological replay, immediate fail-closed suppression, and prospective recovery. This document distinguishes receipt validity, functional acceptance, renderability, statistical evidence, and public-release verification; none is substituted for another.

1 Technical summary and claim decision

Study boundary. V15 concerns one exact candidate, three frozen repositories, six counterbalanced matched pairs per repository, and one provider/model/effort configuration. Cross-fixture pooling is descriptive because the tasks are heterogeneous. Model-panel grades are not independent human-practitioner evidence. Dollar cost is not estimated because billing data were not measured.

Current decision. No confirmatory quality, efficiency, or superiority claim is authorized while generated analysis artifacts are absent.

Evidence completion. Pending receipt-ledger validation.

Primary gate. Pending blinded grading and executed analysis.

Release gate. Memi 2.7.4 publication is not established by this draft.

Pending generated artifact. The executed notebook must state the task-level decisions, exclusions, and exact scope of any claim. (expected at `generated/tex/technical-summary.tex`).

An affirmative task-level quality statement is allowed only if the one-sided 95% paired-bootstrap lower bound for

$$\Delta_q = q_{\text{Memi}} - q_{\text{baseline}} \quad (1)$$

is strictly greater than the preregistered -5 point non-inferiority margin. Passing that gate establishes non-inferiority for the named task; it does not, by itself, establish superiority, lower cost, or generality.

2 Exact 2.7.3 release reconstruction

The candidate under test is the published npm tarball, not a mutable checkout. Its SHA-256 is `b2b56189f45b69a3cec41220ae3e9d50d6ecfb682aee5a64c1fe0ce0d0ec3e1e`; its npm SHA-1 is `f00b05e06fae353322b119243f02a370423a7a88`. The canonical tag source is `2e29c5c656ac34242369eac9840838619ad113e1`. Candidate provenance also records npm integrity, registry attestation, publish workflow run `30684627316`, and publication time.

The release ledger supports a narrow conclusion: relative to the V14 source, 2.7.3 changed release metadata, guidance, and adapter-test stabilization rather than the benchmarked runtime paths. V15 nevertheless executes the exact 2.7.3 artifact, so this reconstruction is not used as a proxy for execution.

3 V14 evidence audit

V14 remains feasibility evidence for 2.7.2. Its freeze and plan hashes are, respectively, `48cb8db737bbebbaea534ffab10ec87d82815f11b310db7b8e8c6db741237759` and `6bb2d2b6f298b68c9464dfd773d7b9b4df85dd9822c035f4fe42d91af1708fb6`. The ledger verifies 12 cells and six pairs over Buzzr and Paraform, with three content-addressed evaluation batches. V14 contains neither Nate/SwiftUI cells nor a rendered blinded-quality panel. It is not relabeled, pooled into V15 inference, or used to fill V15 missing values.

This separation replaces the stale five-pair 2.7.0 analysis path with V15 as the only source for exact-2.7.3 confirmatory claims while retaining V14 for audit history.

4 V15 design and data-quality controls

The frozen design has 18 matched pairs and 36 serialized cells. Every pair shares model `gpt-5.6-luna`, low reasoning, a 450,000-input-token ceiling, task contract, and verification rules. Condition order is counterbalanced with seed 2715. Each cell starts from a clean clone at a pinned revision, uses an immutable fixture, runs isolated post-patch verification, and writes content-addressed evidence. Xcode and Simulator access is exclusive.

Receipt gate. Admission requires a unique frozen trial identity and all registered artifacts, including patch, route, usage, tools, environment, isolated verification, evidence manifest, and run receipt. Hash, fixture, condition, timestamp, candidate, and environment mismatches fail closed. Duplicate or corrupt receipts are errors, not replicates.

Missingness and exclusions. No value is imputed. A cell may retain its functional and resource outcome while being excluded

Table 1: Reconstructed 2.7.3 provenance chain. Abbreviated hashes are display labels; full values remain in `release-provenance.json`.

Commit	Audited role
0a4228bd	Sealed prospective workflow harness.
38b1da10	Workflow-adapter test stabilization only.
b6beb464	2.7.3 candidate declaration.
a642d6c4	Release-surface preparation.
2e29c5c	Canonical 2.7.3 tag source.
9a47f390	npm provenance record.
be69d041	Published release truth and website artifact.
f9cb9987	Main-branch provenance anchor.

Table 2: Frozen V15 task strata.

Task	Platform	Pairs
Buzzr unread badge	Expo	6
Paraform command menu	Web	6
Nate reduced motion	SwiftUI	6

from rendered grading when the exclusion ledger records, for example, a build failure, absent renderable patch, or budget breach. Provider failures and retries remain outcomes. The first orchestration deviation separated the exact 2.7.3 runtime from the hash-pinned 2.7.4 admission harness without changing candidate output. The second retained a stalled partial attempt as a provider/harness retry and used only a later complete receipt for the trial outcome.

Blinding. Three model graders independently score anonymized rendered evidence on task and interaction correctness (25), accessibility (20), visual hierarchy (15), design-system consistency (15), responsive/adaptive behavior (15), and implementation quality (10). Panel medians, absolute disagreement, and ICC(2,1), where estimable, are reported. The label remains *model-graded*.

5 Functional and design-quality results

Pending generated artifact. A validated three-grader panel and complete receipt ledger are required before this section may contain numerical results. (expected at `generated/tex/functional-quality-results.tex`).

Pending generated artifact. The TeX task summary must be generated from `generated/tables/primary_task_summary.csv`, not transcribed manually. (expected at `generated/tex/primary-task-summary.tex`).

The analysis reports task-level paired means and medians, exact sign tests, 10,000-sample paired-bootstrap intervals, and leave-one-pair-out sensitivity. An excluded rendered pair reduces the quality-analysis sample for that task but does not erase its prespecified functional or resource outcome. All exclusions must appear in `exclusions.json` with no imputation.

6 Resource-efficiency results

Pending generated artifact. Generate paired input/output/reasoning-token, wall-time, tool-call, retry, and provider-failure summaries from immutable run receipts. (expected at `generated/tex/resource-efficiency-results.tex`).

For each metric, the raw paired difference is Memi minus baseline. Exact sign tests are oriented so that “better” means higher functional acceptance and lower defects, tokens, time, calls, retries, or provider failures. Holm adjustment is applied across task-level secondary tests. Pooled values are descriptive. No token count is converted to US dollars.

7 Time-forward skill-fitness backtest

The 2.7.4 router indexes evidence by the exact tuple

$$R = (\text{task class, repository fingerprint, provider, model, effort, skill ID, skill content hash}). \quad (2)$$

Evidence from any other tuple is ineligible. Fitness evidence v2 adds immutable blinded-quality evidence and its SHA-256 while preserving v1 loading. A v1 negative event may suppress a route, but automation-only v1 evidence may not promote or recover one.

The chronological replay consumes events only when $t_{\text{event}} \leq t_{\text{as-of}}$. It suppresses immediately after a quality regression or after a joint catastrophic regression of at least 50% more tokens and 25% more latency. Recovery requires three later healthy, prospective, exact-match pairs. An unrecovered route remains disabled, with repository-only discovery.

Pending generated artifact. Insert the stable JSON replay result, route transitions, and final exact-route states. (expected at `generated/tex/fitness-backtest-results.tex`).

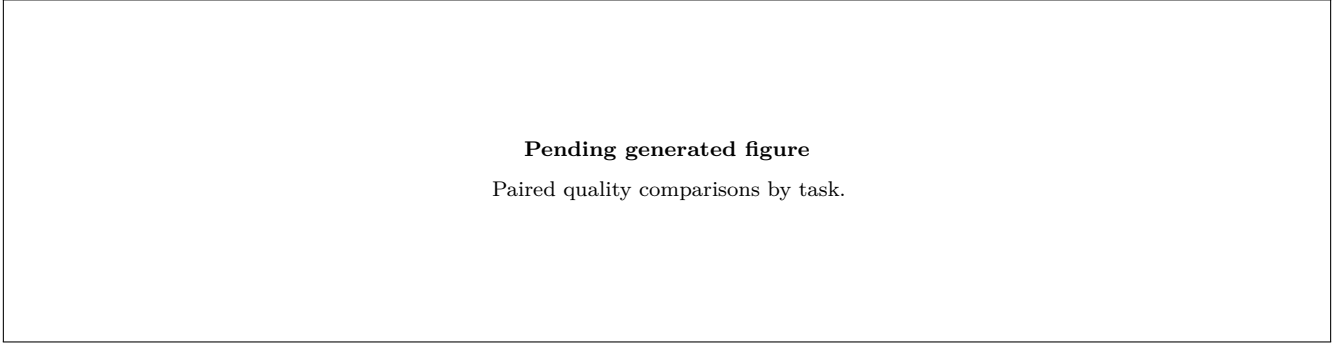


Figure 1: Paired design-quality comparison by frozen task. Scores are blinded panel medians on the 100-point rubric; deltas are Memi minus baseline.

Interpretation.

Pending generated artifact. State which task-level non-inferiority gates passed and distinguish non-inferiority from superiority. (expected at `generated/tex/quality-figure-interpretation.tex`).

Limitation. Only renderable, ledger-admitted evidence enters blinded quality grading. Model graders are not independent practitioners, excluded pairs are not imputed, and cross-task aggregation is descriptive because task semantics and platforms differ.

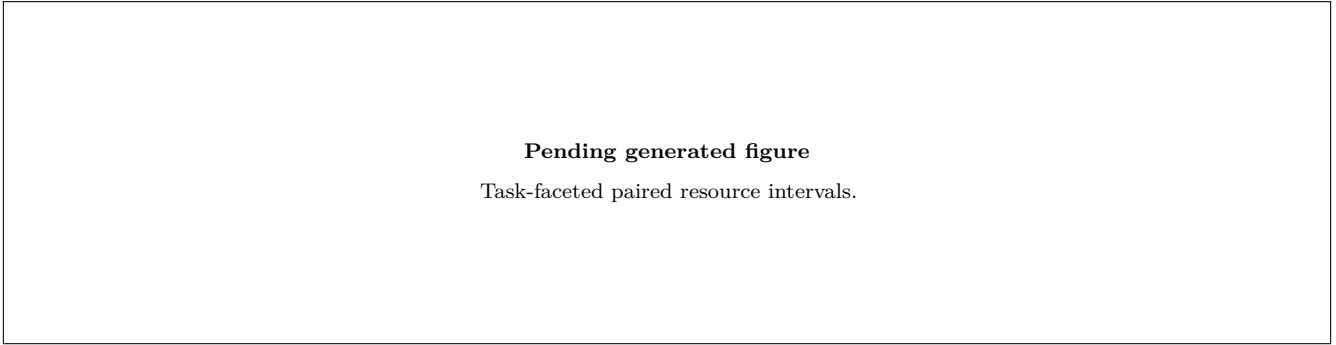


Figure 2: Task-faceted paired resource intervals for the prespecified secondary outcomes. Direction and units must be displayed on each facet.

Interpretation.

Pending generated artifact. Describe the task-specific direction, interval, and Holm-adjusted evidence without converting tokens to cost. (expected at `generated/tex/resource-figure-interpretation.tex`).

Limitation. Wall time includes provider and tool variability on one serialized workstation; token measures are provider-reported usage rather than billing records. With six pairs per task, intervals are expected to be wide and secondary tests have limited power.

8 Website/frontend production audit

The website is audited from a clean checkout. Product-source review excludes `.vercel`, build output, generated community examples, and other non-product sources from the design score. The verification matrix includes unit tests, Astro production build, Chromium journeys, axe checks in light and dark modes, keyboard/focus behavior, reduced motion, responsive states, size budgets, Lighthouse, and production smoke tests. Screenshot review covers empty/error states and available light/dark variants.

Only observed defects are eligible for repair. Infrastructure that prevents a test from executing is reported separately from a product defect, and a green local test is not treated as deployment proof.

Pending generated artifact. Insert source-scoped before/after scores, verified fixes, remaining findings, Lighthouse/axe output, and production smoke evidence. (expected at `generated/tex/website-audit-results.tex`).

9 2.7.4 remediation and release gates

The product remediation is intentionally narrow: exact route isolation, content-hash isolation, v1-compatible v2 quality evidence, stable chronological JSON replay, immediate suppression, prospective recovery, corrupt/duplicate event rejection, and repository-only fallback. Tests are required for each policy boundary before implementation. No route is force-enabled to make a release result appear favorable.

Pending generated artifact. Insert reviewed 2.7.4 commit identities, complete cross-platform test evidence, security review, and the final state of each measured route. (expected at `generated/tex/remediation-results.tex`).

Pending generated artifact. Insert npm trusted-publisher OIDC, fresh install, GitHub tag/assets/checksums, Action v2, Homebrew, GHCR, MCP Registry, website PDF, and public gate evidence. The public gate must report `failures: []` and `parityEligible: true`. (expected at `generated/tex/release-gates.tex`).

Publication is established only after each channel resolves to 2.7.4 and the PDF checksum attached to the GitHub release matches this artifact. A source commit, local package, or passing CI job is not a substitute for registry and production checks.

10 Limitations, falsification criteria, and next study

Limitations. The design uses three intentionally different fixtures, one provider/model/effort setting, six pairs per task, and one Apple-silicon workstation. Model grading can be systematic yet share model-family biases. Provider failures, input ceilings, compile failures, and renderability exclusions reduce the evidence available for some outcomes. Serial execution controls machine contention but does not estimate multi-user throughput. The study does not measure billing cost, maintainability after

long-term use, independent practitioner preference, or general performance on unregistered repositories.

Falsification criteria. Any of the following invalidates an affected claim: candidate checksum mismatch; duplicate or non-frozen trial identity; fixture mutation; condition leakage to graders; unverifiable evidence hash; look-ahead in policy replay; unreported exclusion; manual headline number that differs from the executed notebook; a task-level non-inferiority lower bound at or below -5 ; a critical accessibility or security finding; a LaTeX build with undefined references or overfull boxes; or a release channel that does not resolve to the audited bytes.

Next study. A stronger follow-up should preregister more repositories per platform, independent human practitioners, multiple provider/model/effort strata, measured billing records, repeated hardware environments, and a longer prospective recovery horizon. New evidence must remain segregated by the exact route tuple and content hash.

Reproducibility and artifact contract

The executed source of analysis is `analysis.ipynb`; all headline numbers must independently reproduce from immutable receipts. Machine-readable outputs are expected at:

- `generated/tables/primary_pair_level.csv`, `primary_task_summary.csv`, and `leave_one_pair_out.csv`;
- `generated/tables/secondary_task_tests.csv`, `pooled_descriptive.csv`, and `grader_reliability.csv`;
- `generated/tables/analysis_summary.json`; and
- the TeX and figure hooks enumerated in `report-hooks.tex`.

The protocol, plan, environment, rubric, candidate/release provenance, exclusion/deviation ledgers, receipt ledger, grading records, tables, figures, notebook, TeX source, and final PDF require a checksum inventory before public release.

Pending generated figure
Chronological, no-look-ahead routing-policy replay.

Figure 3: Chronological, no-look-ahead replay of the exact-match routing policy. State changes are derived only from evidence available at each as-of time.

Interpretation.

Pending generated artifact. Name each suppression or recovery transition and identify the exact evidence event that caused it. (expected at `generated/tex/backtest-figure-interpretation.tex`).

Limitation. A deterministic backtest validates policy mechanics and guards against look-ahead leakage; it does not establish that the historical event stream is representative of future repositories, providers, models, or skill revisions.